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Article *in* Journal of Structural Engineering · May 1992

DOI: 10.1061/(ASCE)0733-9445(1992)118:5(1270)

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Prestressed FRP Sheets as External Reinforcement of Wood Members

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Abstract: High strength, low weight, corrosion resistance, and electromagnetic neutrality make fiber-reinforced plastic (FRP) a suitable candidate in many structural applications, including rehabilitation and strengthening as well as the development of new wood members. Advanced forms of reinforced wood construction can enable contemporary wood structures to play an even greater role in today's construction. In this work, the writers establish a novel technique for reinforcing wood members involving external bonding of pretensioned FRP sheets on their tension zones. An analytical model for the maximum initial pretension is verified with tests on carbon/epoxy-prestressed wood beams. Additional studies, both analytical and experimental of the flexural behavior of wood beams reinforced with prestressed carbon/epoxy FRP sheets demonstrate the superior performance of the hybrid system and emphasize its favorable strength, stiffness, and ductility characteristics. Finally, a methodology is described for the selection of composite material dimensions and initial prestressing to maximize structural performance.

Introduction and Background

Wood is the oldest and one of the most widely used structural materials. Improved design methods and advanced forms of wood construction, involving the use of reinforcement to enhance the mechanical properties of wood members, can enable contemporary and advanced forms of large wood structures (e.g., long-span bridges) to be at least as reliable and economically competitive as those constructed from other construction materials, such as concrete, steel, and plastics.

Several attempts to reinforce wood elements have been reported in the literature. A very common method to reinforce wood has been to use steel or aluminum plates. Sliker (1962) reinforced laminated wood beams with aluminum strips placed between selected laminations. Mark (1961, 1963) reinforced wood sections by bonding aluminum facings on the outsides of the beams. Steel plates have been placed between laminations both vertically and horizontally (Borgin et al. 1968; Stern and Kumar 1973; Coleman and Hurst 1974; Hoyle 1975). Reinforcement of wood with both square and round cross-section rods was examined by several investigators (e.g., Lantos 1970; Dziuba 1985; Bulleit et al. 1989). High-strength steel wire embedded in an epoxy matrix has been used to replace tension laminations of wood beams (Krueger 1973; Krueger and Eddy 1974; Krueger and Sandberg 1974; Kobetz and Krueger 1976). Another method of reinforcing wood is to use prestressed steel reinforcement. Horizontally laminated wood beams were prestressed by posttensioning unbonded high-strength steel strands in the tension zone of the beams (Bohannan 1962). One unique attempt has been made to prestress glu-lam using pretensioned steel plates bonded on the

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Note. Discussion open until October 1, 1992. To extend the closing date one month, a written request must be filed with the ASCE Manager of Journals. The manuscript for this paper was submitted for review and possible publication on May 12, 1991. This paper is part of the *Journal of Structural Engineering*, Vol. 118, No. 5, May, 1992. ©ASCE, ISSN 0733-9445/92/0005-1270/$1.00 + $.15 per page. Paper No. 1843.

J. Struct. Eng. 1992.118:1270-1284.

tension face with epoxy adhesive (Peterson 1965). A main disadvantage of the various reinforcing methods involving metal plates, particularly when used in exposed wood constructions, is the high risk of reinforcement corrosion.

Glass fiber-reinforced plastics (GFRP) have been used as faces of wood-core sandwich beams (e.g., Biblis 1965; Boehme and Schultz 1974). Theakston (1965) reported increases in strength and stiffness of wood members when they were wrapped in GFRP or when the composite was placed between horizontal laminations. A unique attempt was made to incorporate prestressed GFRP strands into structural members made from wood flakes (Saucier and Holman 1975).

In a recent work, Plevris and Triantafillou (1992) provided a comprehensive study of the short-term flexural behavior of wood beams and beam columns reinforced with unidirectional fiber-reinforced plastic (FRP) sheets, bonded on the members' tension face only. Their work shows that even extremely thin FRP sheets as external reinforcement of wood structures offer several advantages, increasing the strength, stiffness and ductility characteristics of members considerably. The study was verified experimentally by testing wood beams and beam columns, reinforced with unidirectional carbon/epoxy FRP (CFRP) sheets.

Composite reinforcement in the form of unidirectional plates bonded on the tension face of wood members appears to be a promising method, for both building new wood structures and strengthening deteriorated ones. The method's main advantages can be easily realized considering that the reinforcement consists of strong, corrosion-immune, lightweight, and electromagnetically neutral materials, placed at a certain location where they can perform best. This method can be further extended when the FRP sheets are prestressed before being applied on the wood surface (Triantafillou and Deskovic 1991). Prestressing may be justified when FRP material cost becomes an issue (e.g., when CFRP composites are used). A pretensioned laminate of given thickness and a thicker but unstressed one can be equally efficient as far as enhancement of a member's mechanical properties is concerned.

An analytical model describing the maximum achievable prestress level, so that the FRP-prestressed system does not fail near the two ends upon releasing the prestressing force, has been developed by Triantafillou and Deskovic (1991). The model accounts for either adhesive shear failure or failure of the beam material, when the beam is made of concrete. In this study, the model is extended to describe failure of wood beams: it is also verified through a series of experiments. Finally, the flexural behavior of wood beams prestressed with externally bonded FRP sheets is discussed based on analytical and experimental results.

Maximum Initial Prestress

Central to the development of the concept of reinforcing wood structures with prestressed FRP sheets is the calculation and verification of the maximum initial prestress so that, upon releasing it, the member does not fail.

Analysis

Consider the beam illustrated in Fig. 1; it has length l, height h, and width b. The FRP sheet has thickness t and width b_{fc}; the adhesive has thickness d. The material properties are as follows: E_w = Young's modulus

J. Struct. Eng. 1992.118:1270-1284.

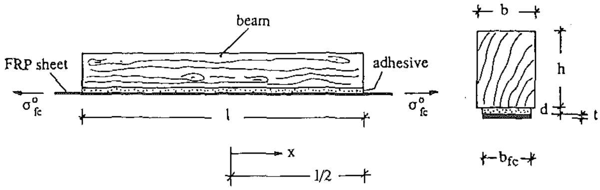

FIG. 1. Wood Beam Prestressed with Epoxy-Bonded Fiber-Reinforced Plastic Sheet

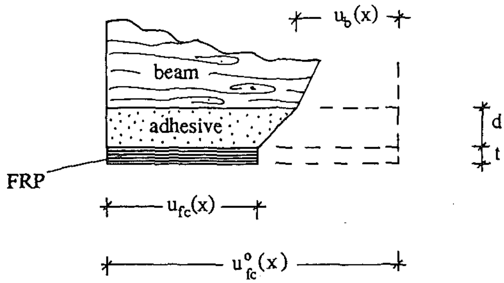

FIG. 2. Deformation Pattern of FRP-Adhesive-Wood System: Dashed Lines = Configuration during Pretensioning; Solid Lines = Configuration After Pretension Is Released

of wood; G_w = shear modulus of wood; G_a = shear modulus of adhesive; and E_{fc} = Young's modulus of fiber-composite sheet. When the initial pretension level σ_{fc}^0 is released, the normal stress in the FRP drops to $\sigma_{fc}(x)$ and a prestress $\sigma_p(x)$ develops at the bottom fiber of the beam.

Based on the displacement pattern of the system FRP-adhesive-wood at a distance x from the midspan before and after the pretension is released (see Fig. 2), considering stress equilibrium and displacement compatibility, and assuming linear elastic material behavior, the method of Triantafillou and Deskovic (1991) gives the shear stress next to the beam-adhesive interface as

$$\tau = Ae^{\omega x} + Be^{-\omega x} \quad (1)$$

where A and B = constants; ω is given by

$$\omega = \sqrt{\frac{G_a}{dt'}\left(\frac{1}{E_{fc}} + \frac{\alpha}{E_w}\right)}, \qquad t' = t\,\frac{b_{fc}}{b} \quad (2)$$

and α = a parameter relating the prestress at the bottom fiber of the beam, σ_p, to the tensile stress in the fiber composite, σ_{fc} ($\sigma_p = -\alpha\sigma_{fc}$). α depends on the cross-section shape and equals $4t'/h$ for rectangular cross sections. The following notation is used in Fig. 2: $u_{fc}^0(x)$ = initial extension of the FRP; $u_{fc}(x)$ = extension of the FRP shortly after the pretension is released; and $u_b(x)$ = shortening of the wood at the bottom fiber.

A typical shear stress-strain relationship for wood is given in Fig. 3. Fracture of the material in shear occurs after it yields plastically (e.g., Bodig and Jayne 1982). The true shear stress-strain relationship can be approximated by a bilinear curve describing two regimes of behavior, one linear

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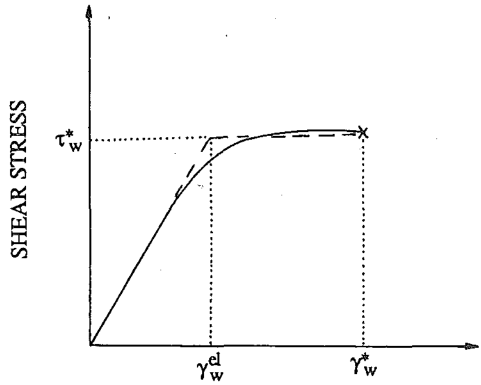


FIG. 3. Exact and Idealized Shear Stress-Strain Relationship for Wood

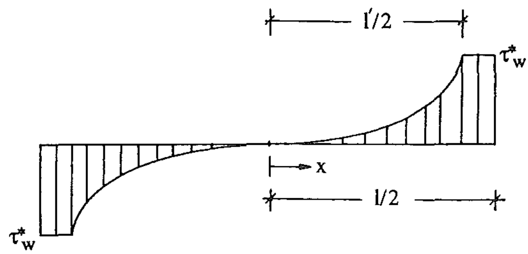


FIG. 4. Shear Stress Distribution in Wood Beam, Next to Adhesive-Wood Interface

elastic and the other perfectly plastic (Fig. 3, dashed lines), indicating that the shear stress distribution just before failure due to pretension release can be approximated by that shown in Fig. 4.

Eq. (1) along with the boundary conditions

$$\tau(x = 0) = 0 \qquad (3)$$

$$\gamma\left(x = \frac{l'}{2}\right) = \frac{\tau\left(x = \frac{l'}{2}\right)}{G_w} = \gamma_w^{el} \qquad (4)$$

give the shear strain distribution in the wood next to the interface as follows:

$$\gamma = \frac{\gamma_w^{el}}{\sinh \frac{\omega l'}{2}} \sinh \omega x, \qquad 0 \le x \le \frac{l'}{2} \qquad (5)$$

Next, assuming that the strain distribution described by (5) is valid in the plastic zone, too (i.e., for $0 \le x \le l/2$), and from the condition $\gamma(x = l/2) = \gamma_w^*$, the length of the elastic zone is calculated as

$$l' = \frac{2\ln\left(\frac{\beta + \sqrt{\beta^2 + 4}}{2}\right)}{\omega}, \qquad \beta = \frac{2\gamma_w^{el}}{\gamma_w^*}\sinh\frac{\omega l}{2} \quad \text{(6)}$$

Finally, stress equilibrium and slope continuity of the normal stresses σ_{fc} at $x = l'/2$ give the initial prestress σ_{fc}^o, which, when released, will just result in failure of the system by shearing of the wood as

$$\sigma_{fc}^o = E_{fc} d \gamma_w^{el} \omega \left[\coth\frac{\omega l'}{2} + \frac{\omega(l - l')}{2}\right] \quad \text{(7)}$$

A similar derivation can be found in more detail for the case of adhesive shear failure in Triantafillou and Deskovic (1991).

Experimental Procedure

The validity of the analytical solution for the maximum pretension of FRP sheets bonded on the tension face of wood beams was verified from experiments on three specimens 800 mm long and 45 mm wide. Specimens A and B had a cross-section height of 60 mm, while specimen C had a height of 80 mm. The Young's modulus was obtained by loading the beams elastically in three-point bending; and the shear stress-strain response was characterized by performing the double shear test on three specimens of the same material (European beech). The average value of the Young's modulus was $E_w = 13.5$ GPa. The double shear test resulted in the following values: ultimate elastic shear strain $\gamma_w^{el} = 0.012$; and shear strain at failure $\gamma_w^* = 0.039$.

Thin carbon/epoxy sheets were used in the experimental program, consisting of unidirectional carbon fibers at a volume fraction of about 60% bonded together with an epoxy matrix. Uniaxial testing of the laminates gave an average elastic modulus $E_{fc} = 115$ GPa and a strain at failure $\varepsilon_{fc}^* = 0.0122$. The CFRP sheet bonded on specimen A had a thickness of 0.75 mm, while the sheets bonded on specimens B and C had a thickness of 0.50 mm. All three laminates had a width of 44 mm.

The epoxy adhesive employed in the test program was chosen such that viscoelastic phenomena were minimized; it contained quartz powder and sand as filler materials to improve the creep behavior and reduce cost. The shear modulus of the adhesive was measured by performing the torsion pendulum test on 100 × 10 × 1-mm thin prisms; it was found to be 2.8 GPa at 20°C.

The problem to be solved when implementing the prestressing method in practice is clamping of the two FRP ends. In this study, this was achieved through the use of two pairs of steel plates with gradually decreasing thickness to minimize stress concentrations. Each pair of plates was bonded on one end of the CFRP sheet with a high-performance epoxy adhesive, cured in a hydraulic press at constant pressure (10 atm) and temperature 65°C) for 2 hours. The steel plates were sandblasted and the composite sheet was roughened with sandpaper before the adhesive was applied over them. The double-lap adhesive joints had a length of 100 mm. The two steel-CFRP lapped joints were hinge-connected to a fixed load cell at one end and a hydraulic jack at the other. The hydraulic jack and load cell were mounted on the two ends of a stiff steel beam used to support the wood beams.

Special consideration was given to the surface preparation before bonding the composite sheets on the wood face. Vacuum cleaning was employed to remove the dust and loose particles from the wood surface. The CFRP surface was roughened with sandpaper and subsequently cleaned with acetone, shortly before the adhesive was spread over it.

Following surface preparation, each CFRP laminate was pretensioned up to the desired load value, the adhesive was applied on both the wood and the composite material surfaces and the beam was placed on top of the sheet. Weights increased the lateral pressure to about 0.06 MPa. A continuous support was provided under the CFRP with a soft wood plank. After the adhesive was hardened for about three days, the weights and wood support were removed. The pretension force was then gradually released from one end with the hydraulic jack, until the first visible horizontal crack appeared at the ends of the beam right above the adhesive-wood interface, while the load was recorded. The difference between the load corresponding to first fracture (failure load) and the initial prestress gave the maximum achievable pretension level so that failure of the system would not initiate near the two ends.

Results

The results obtained from the experiments on prestressing the wood members are summarized in Table 1. A typical failure pattern at the end of one member while the pretension was released is illustrated in Fig. 5(*a*); the final state of the beams after the force was fully released is illustrated in Fig. 5(*b*). It was observed that the initial horizontal shear crack propagated through the wood layer just above the adhesive-wood interface, as expected. A comparison between the analytical and experimental maximum prestress level for the beams tested is given in Fig. 6 as a function of the adhesive thickness, which was estimated to be in the range 0.75–1.25 mm; a good agreement between theory and experiment can be noticed. If the average adhesive thickness is defined in the middle of the expected range, that is $d = 1$ mm, the average error is 11%, while the maximum error is 17.5% corresponding to beam A. It is concluded that the results obtained in this study, although rather limited, add significant confidence to the analytical model for the determination of the maximum pretension [(7)].

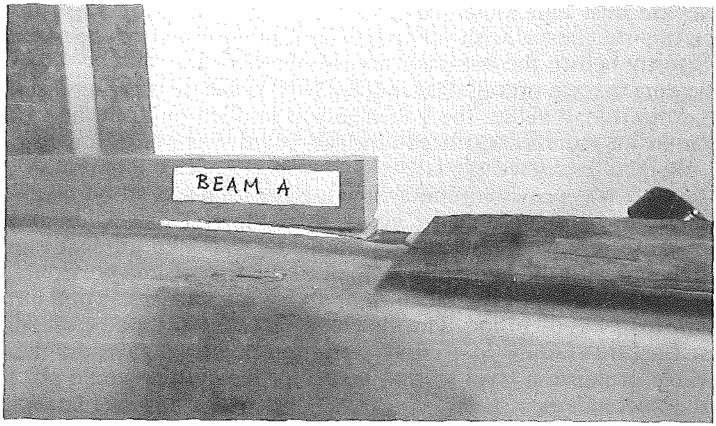
FRP-Prestressed Wood: Flexural Behavior

Analysis

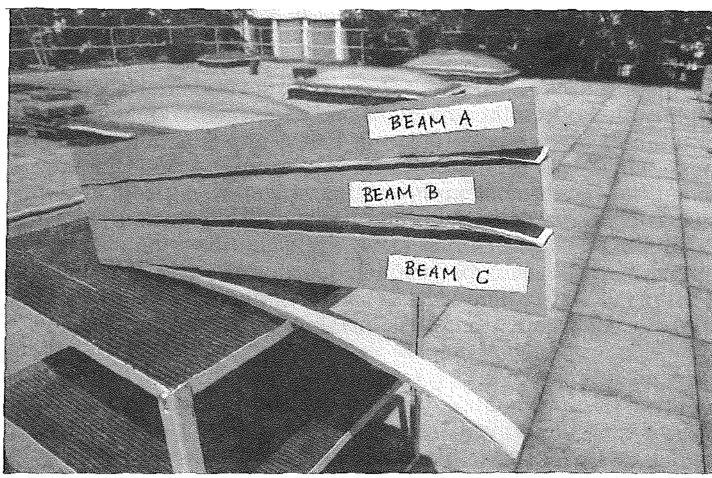
Prestressing with externally epoxy-bonded FRP sheets can be thought of as both a strengthening technique and a method of reinforcing wood structures in new constructions. The flexural behavior of wood beams in both cases can be determined by using concepts of classical flexural theory along with the strain compatibility method, as they apply to the analysis of prestressed cross sections. For instance, a proper design requires that at the

TABLE 1. Test Results on CFRP-Prestressed Wood Members

Beam number (1)	Initial prestress (MPa) (2)	Prestress at failure (MPa) (3)	Maximum achievable prestress, σ_{fc}^{o} (MPa) (4)
A	883	93	790
B	822	22	800
C	932	114	818



(a)



(b)

FIG. 5. Failed CFRP-Wood Beams: (*a*) Failure Initiation Near End; and (*b*) Final State of Beams

ultimate flexural capacity, wood compressive yield precedes tensile fracture of either the extreme wood tensile fiber or the composite sheet.

The model of wood uniaxial behavior used here is the one proposed by Bazan (1980) and modified by Buchanan (1990) (Fig. 7). The same model gave satisfactory results for the prediction of the moment capacity of wood members reinforced with externally bonded nonprestressed CFRP sheets (Plevris and Triantafillou 1991). The stress distribution effect at a cross section can be quantified by considering the following relationship between the tensile strength f_t and the extreme wood fiber tensile stress at failure, f_m (Buchanan 1986):

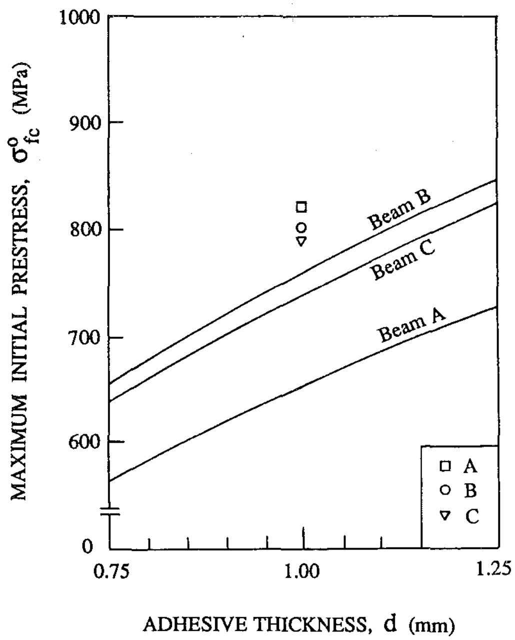


FIG. 6. Comparison between Analysis and Experiments for Maximum Initial Prestress

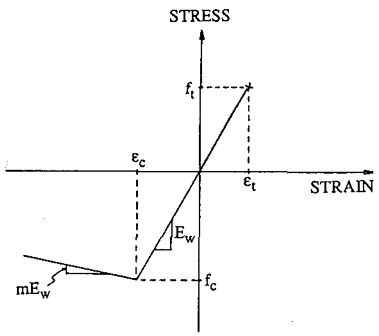


FIG. 7. Idealized Uniaxial Stress-Strain Relationship for Wood

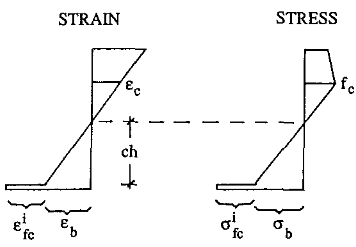


FIG. 8. Strain and Stress Profile at Failure of FRP-Prestressed Wood Cross Section

$$f_m = \left(\frac{c}{1 + k}\right)^{-1/k} f_t \quad \text{(8)}$$

where c = the ratio of the depth of the tensile zone, measured from the extreme tensile fiber, to the total section height; and k = the stress distribution parameter which reflects the variability in strength properties within

J. Struct. Eng. 1992.118:1270-1284.

the depth of the member (low k indicates high variability). For specimens with low variability in strength properties, $f_m \approx f_t$.

Referring to Fig. 8, the normalized ultimate bending moment at failure can be calculated as follows:

$$\frac{M_u}{bh^2E_w} = \max\left(C_2\left\{1 - \left[1 - \left(\frac{\varepsilon_c}{\varepsilon_b} + 1\right)c\right]\frac{\dfrac{3+m}{m}\dfrac{c}{c-1}\dfrac{\varepsilon_c}{\varepsilon_b} + 1}{\dfrac{2+m}{m}\dfrac{c}{c-1}\dfrac{\varepsilon_c}{\varepsilon_b} + 1}\right\} + C_1 c\left(1 + \frac{2}{3}\frac{\varepsilon_c}{\varepsilon_b}\right) - \frac{Tc}{3}\right) \quad (9)$$

where

$$T = \frac{c}{2}\varepsilon_b \quad (10a)$$

$$C_1 = \frac{c}{2}\frac{\varepsilon_c^2}{\varepsilon_b} \quad (10b)$$

$$C_2 = \frac{1}{2}\left[1 - c\left(\frac{\varepsilon_c}{\varepsilon_b} + 1\right)\right]\left[(2+m)\varepsilon_c + m\left(1 - \frac{1}{c}\right)\varepsilon_b\right] \quad (10c)$$

and c is given by solving

$$(1+m)\left(1 + \frac{\varepsilon_c}{\varepsilon_b}\right)^2 c^2 + \left[2\rho_{fc}\frac{E_{fc}}{E_w}\frac{(\varepsilon_{fc}^i + \varepsilon_b)}{\varepsilon_b} - 2(1+m)\frac{\varepsilon_c}{\varepsilon_b} - 2m\right] \cdot c + m = 0 \quad (11)$$

under the conditions $(1 - c)\varepsilon_b/c \geq \varepsilon_c$ (wood yield); $\varepsilon_b + \varepsilon_{fc}^i \leq \varepsilon_{fc}^*$ (FRP fracture); and $\varepsilon_b \leq f_m/E_w$ (wood fracture). ε_{fc}^i is the prestress applied initially in the FRP sheet and $\rho_{fc} = b_{fc}t/bh$ is the area fraction for the fiber composite.

For each particular set of values for the FRP area fraction and the initial prestress ($E_{fc}\varepsilon_{fc}^i$), the relationship between the bending moment and the extreme tensile fiber strain (ε_b) can be established from (9), and the maximum bending moment can thus be calculated. A computer program was developed to facilitate these calculations and produce the diagram of Fig. 9. All points in Fig. 9 correspond to ultimate moment capacities achieved before either the FRP or the wood fracture; except for those of curve B-C. The figure, based on the material properties of Table 2, shows the effect on the member's flexural strength of both the composite area fraction and the initial prestress (expressed as a percentage of the ultimate prestress dictated by the FRP strength).

Prestressing increases the ultimate bending capacity of the member to a significant degree, provided that for a certain range of FRP area fractions (low) the initial pretension is not too high. In other words, each ρ_{fc} is associated with a critical value of initial FRP strain, which results in a moment capacity just equal to the capacity of the nonprestressed member.

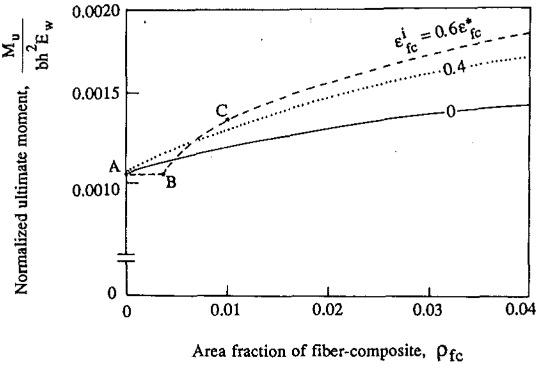


FIG. 9. Normalized Ultimate Bending-Moment as Function of Fiber-Composite Area Fraction for Various Initial Pretension Levels (A-B = Strength of Unreinforced Wood Controls; B-C = FRP Fractures)

TABLE 2. Material Properties

Property (1)	Wood (2)	CFRP (3)
E_w(GPa)	13.5	—
m	0.03	—
f_c(MPa)	72.9	—
f_t(MPa)	175.5	—
k	100.0	—
E_{fc}(GPa)	—	115
ε^*_{fc}	—	0.0122

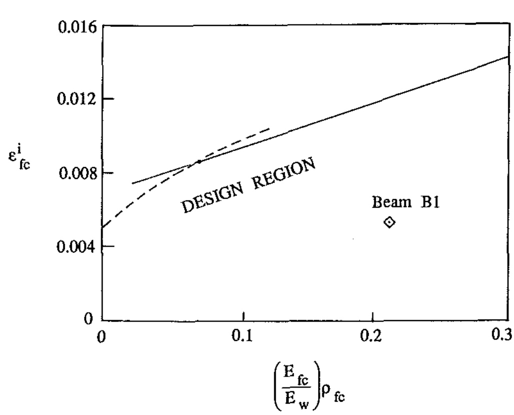


FIG. 10. Limiting Pretension Strain as Function of FRP Area Fraction

J. Struct. Eng. 1992.118:1270-1284.

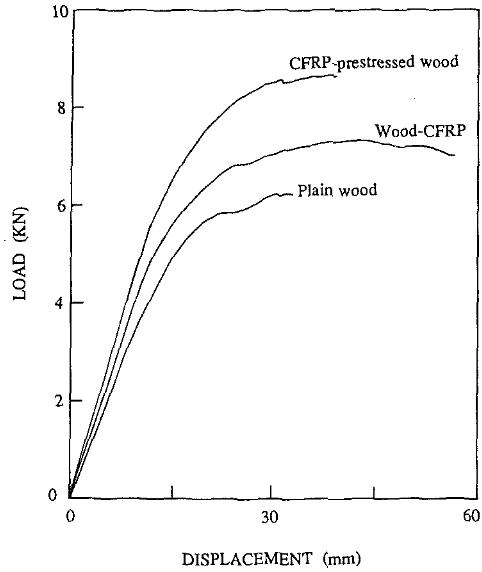


FIG. 11. Three-Point Bending Load versus Crosshead Displacement Curves for Plain, Reinforced, and Prestressed Wood

This condition is described by the solid line of Fig. 10. Another limit to ε_{fc}^{i} must be established to ensure that wood yield initiates before the cross section fails. From the analysis of cross sections, it can be shown that wood yield precedes FRP fracture when

$$\varepsilon_{fc}^{i} \leq \left(1 + 2\rho_{fc}\frac{E_{fc}}{E_{w}}\right)\varepsilon_{fc}^{*} - \varepsilon_{c} \quad \text{(12)}$$

under the condition $\varepsilon_{fc}^{i} \geq \varepsilon_{fc}^{*} - f_m/E_w$, while wood yield precedes wood tensile fracture when

$$\varepsilon_{fc}^{i} \leq \frac{(f_c - f_m)}{2\rho_{fc}E_{fc}} - \frac{f_m}{E_w} \quad \text{(13)}$$

under the condition $\varepsilon_{fc}^{i} \leq \varepsilon_{fc}^{*} - f_m/E_w$. The limit on ε_{fc}^{i} imposed by inequalities in (12) and (13) is described by the dashed line of Fig. 10. [Eq. (12) controls for the material properties of Table 2.]

Experimental Method and Results

Three 800-mm-long, 30-mm-wide, and 40-mm-deep wood beams similar to those used to verify the analytical model of (7) (E_w = 13.5 GPa) were tested in three-point bending to study the effect of prestressed FRP sheets on their flexural behavior. The wood specimens were cut from European beech lumber with the strong material direction parallel to the member axis. Special efforts were made to obtain uniform and "clean" specimens without knots, thus minimizing the uncertainties associated with the variability in mechanical properties. The yield strength and slope of the postyield branch

J. Struct. Eng. 1992.118:1270-1284.

of wood in uniaxial compression were measured by compressing three 25-mm cubes in an Instron testing machine. The following average values were obtained: $f_c = 72.9$ MPa and $m = 0.03$. The experimental procedure and materials (CFRP, epoxy adhesive) described in the previous section were employed to prestress one of the beams. A second beam reinforced with CFRP of the same area fraction was tested without prestressing, and a third beam was tested without reinforcement, as a control specimen. The composite sheets were 30 mm wide and 1 mm thick, giving $\rho_{fc} = 0.025$. The ultimate prestress was calculated from (7) as $\sigma^o_{fc} = 640$ MPa, and the applied prestress was $\sigma^i_{fc} = 620$ MPa.

The three beams were loaded to failure simply supported over a span of 700 mm using an Instron testing machine. Load and midspan displacement were measured during the test and were recorded on an X-Y recorder. The load-displacement curves obtained are given in Fig. 11. When compared to the plain wood specimen, the reinforced but non-prestressed specimen experienced higher-strength, stiffness, and ductility characteristics; the prestressed specimen was even stronger and stiffer, maintaining similar ductility levels. The fact that prestressing in this case increases the strength of the CFRP-wood system can be verified by noting that the combination of composite area fraction, material elastic properties, and initial pretension produce a point inside the design region of Fig. 10 ($\rho_{fc}E_{fc}/E_w = 0.213$, $\varepsilon^i_{fc} = 0.0054$).

Conclusions

The behavior of a new method of reinforcing wood structures was established in this study. The method involves external bonding of pretensioned fiber-reinforced plastic sheets on the tension faces of beams through the use of epoxy adhesives. It combines the advantages offered by composite materials (such as corrosion resistance, low weight, and electromagnetic neutrality) with the high efficiency of external prestressing, to result in the development of hybrid wood-FRP structural members characterized by enhanced mechanical properties. The technique appears to be promising for both rehabilitation/strengthening applications and new constructions. Compared to the method of external bonding of nonprestressed FRP sheets, the technique offers savings in materials, which may be of critical importance, especially when advanced composites, such as CFRP, are selected as prestressing elements. CFRP, on the other hand, offers properties such as high stiffness, high strength, excellent fatigue and creep behavior, and good durability, by far exceeding those of other composite materials.

An analytical model was developed for the prediction of the maximum FRP pretension, so that failure of the wood beam upon releasing the prestress would not occur. Comparison of the model's predictions with experimental results obtained here gave satisfactory agreement, despite the variability associated with the properties of wood structures. This variability, however, was kept to a minimum in this study, by appropriate selection of the wood specimens.

Prestressing with fiber-composite sheets involves the development of a rather simple apparatus. Of particular importance in the prestressing process is the clamping of the two FRP ends and the preparation of the adherent surfaces. In the case of composite materials, clamping of the laminate's end cannot be achieved with mechanical fasteners, because of the material's sensitivity to stress concentrations and in-plane shear stresses. However, clamping can be successfully applied by means of epoxy adhesives. Equally important to the success of the technique is the selection of the adhesive employed to bond the pretensioned laminate on the wood surface; its viscoelastic effects should be minimized.

The flexural behavior of wood beams reinforced with prestressed composites can be established from basic flexural mechanics. For given wood and composite material properties, good designs can be achieved by appropriately selecting the FRP area fraction and the magnitude of the initial pretension. To avoid sudden collapse, it is essential that the members be designed to fail by yielding in the compressive zone first, followed by tensile fracture. Furthermore, it must be ensured that the strength of pretensioned wood beams exceeds that of beams reinforced with nonprestressed sheets. Otherwise, prestressing would not offer any advantages. Thus, a given pretension level corresponds to a minimum FRP area fraction to be used, or a given FRP area fraction corresponds to a maximum pretension. These limiting values can be found from diagrams such as that shown in Fig. 10.

Test results obtained from three-point bending of plain, reinforced, and prestressed wood beams indicate the high potential of the new technique: Appropriate combinations of prestress levels and area fractions of the composite laminates may result in components with excellent strength, stiffness, and ductility characteristics. In addition, introduction of high-reliability elements in the tension zones of wood structures is expected to reduce the high variability associated with their mechanical properties.

Future research should be directed toward establishing the long-term behavior of FRP-prestressed wood beams. In view of the viscoelastic nature of the constituent materials, it is imperative that creep and relaxation be examined in detail before the method can be applied in practice. Moreover, changes in the moisture content of wood could tend to cause shrinkage or swelling. Although these phenomena are not expected to affect the quality of the bond (high-performance rubber-toughened epoxy adhesives, for instance, can accommodate high strains), they may change the level of the prestressing force. Certainly the future study should include members of various grades of lumber within species, different species for which prestressing holds economic promise and different composite materials.

Acknowledgments

Financial support to the project was provided by the National Science Foundation (grant no. MSS-9015542) and the Swiss Federal Laboratories for Materials Testing and Research (EMPA). The writers wish to thank the EMPA director, U. Meier, and M. Deuring, a research engineer and doctoral candidate at EMPA, for their valuable assistance during various stages of the work.

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APPENDIX II. NOTATION

The following symbols are used in this paper:

A = constant coefficient;
a = cross-section parameter;
B = constant coefficient;
b = width of cross section;
b_{fc} = width of fiber-composite sheet;
c = neutral axis depth from tensile fiber as ratio of total beam depth;
d = thickness of adhesive layer;
E_{fc} = Young's modulus of fiber composite;
E_w = Young's modulus of wood;
f_c = compressive strength of wood;

J. Struct. Eng. 1992.118:1270-1284.

f_m = bending strength of wood;
f_t = tensile strength of wood;
G_a = shear modulus of adhesive;
G_w = shear modulus of wood;
h = height of cross section;
k = stress distribution effect parameter;
l = beam length;
l' = length of elastic zone;
M_u = ultimate bending moment;
m = ratio of slopes in compressive stress-strain relationship for wood;
t = thickness of fiber composite;
t' = equivalent thickness of fiber composite;
u_b = horizontal displacement of beam at bottom fiber;
u_{fc} = horizontal displacement of fiber composite after pretension is released;
u_{fc}^o = initial horizontal displacement of fiber composite;
x = distance from midspan;
γ = shear strain;
γ_w^{el} = maximum elastic strain of wood;
γ_w^* = ultimate shear strain of wood;
ε_b = strain of wood bottom fiber;
ε_c = strain corresponding to compressive strength of wood;
ε_{fc}^i = initial pretension strain of fiber composite;
ε_{fc}^* = uniaxial strain of fiber composite at failure;
ε_t = strain corresponding to tensile strength of wood;
ρ_{fc} = area fraction of fiber composite;
σ_b = stress in wood bottom fiber;
σ_{fc} = normal stress in fiber composite;
σ_{fc}^o = maximum initial prestress in fiber composite;
σ_p = prestress at bottom fiber of beam;
τ = shear stress; and
τ_w^* = shear strength of wood.